TUTORIAL 9

Linear Maps and Matrix Representations

MOTIVATION AND GUIDING IDEA

Motivation. A representing matrix is a computational tool that depends on the ordered bases of the domain and codomain. Changing coordinates leaves the map unchanged but can greatly simplify repeated calculations. Always identify the input and output coordinate systems to avoid reversing a change of basis.

NOTATION AND CONVENTIONS

- For ordered bases $B = (u_1, \dots, u_n)$ of U and $C = (v_1, \dots, v_m)$ of V, the notation $[T]_C^B$ denotes an $m \times n$ matrix taking B-coordinates as input and returning C-coordinates.
- $S \circ T$ applies T first and S second; I_V is the identity map on V.
- $P_{B \leftarrow B'} = [I_V]_B^{B'}$ takes $[v]_{B'}$ to $[v]_B$. The arrow indicates the direction of coordinate conversion.
- If B is an ordered basis of $\mathbb{F}^n$ and E is the standard basis, $S_B = [\, b_1 \ \cdots \ b_n \,] = [I]_E^B$ is the basis-column matrix. Hence $[v]_E = S_B [v]_B$.
- $[T]_B$ abbreviates $[T]_B^B$ for an operator $T : V \to V$. Square matrices A, D are similar if $D = P^{-1}AP$ for an invertible P.

DEFINITION 9.1

Let U, V be finite-dimensional vector spaces over the same field, with ordered bases B, C, and let $T : U \to V$ be linear. Its representing matrix is defined by its columns:

$$[T]_C^B = \big[\, [T(u_1)]_C \ \cdots \ [T(u_n)]_C \,\big], \qquad [T(u)]_C = [T]_C^B [u]_B.$$

For another ordered basis B' of the domain, the columns of $P_{B \leftarrow B'}$ are the B-coordinate vectors of the vectors of B'. Its inverse is $P_{B' \leftarrow B}$. If $R : V \to W$ is linear and D is an ordered basis of W, then

$$[R \circ T]_D^B = [R]_D^C [T]_C^B.$$

If T is an isomorphism, $[T^{-1}]_B^C = ([T]_C^B)^{-1}$. For an operator with standard matrix A, the matrix in the ordered basis B is $S_B^{-1} A S_B$. Domain and codomain bases must match at every matrix product.

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MOTIVATION AND GUIDING IDEA

Construct a map from prescribed vector images, check compatibility with linear dependence relations, compute a representing matrix and change bases. The emphasis is on the connection between an abstract map, coordinates and matrix calculations.

9.1 Determining a map from vector images

PROPOSITION 9.1

Let V, W be vector spaces over the same field $\mathbb{F}$, let $B = (v_1, \dots, v_n)$ be an ordered basis of V, and let $w_1, \dots, w_n \in W$ be arbitrary vectors, not necessarily independent. There is a unique linear map $T : V \to W$ with $T(v_i) = w_i$ for every i. Explicitly, if $v = \sum_{i=1}^{n} a_i v_i$, then

$$T(v) = \sum_{i=1}^{n} a_i w_i.$$

The formula is well defined because basis expansions are unique. It is linear because coordinates of sums and scalar multiples are computed entry by entry. Any linear map with the prescribed images must obey this formula, proving uniqueness. In particular, between any two vector spaces over the same field there is at least one linear map: the zero map.

EXERCISE 9.1

Determine whether a linear map T with the following properties exists. If it does, give an explicit formula.

1. $T : \mathbb{R}^3 \to \mathbb{R}^3$ satisfying

$$T\overbrace{\begin{pmatrix}1\\1\\1\end{pmatrix}}^{u_1} = \begin{pmatrix}3\\0\\0\end{pmatrix}, \quad T\overbrace{\begin{pmatrix}2\\-1\\2\end{pmatrix}}^{u_2} = \begin{pmatrix}0\\0\\0\end{pmatrix}, \quad T\overbrace{\begin{pmatrix}1\\0\\0\end{pmatrix}}^{u_3} = \begin{pmatrix}1\\0\\0\end{pmatrix}$$

2. Does there exist a linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$ satisfying

$$T\begin{pmatrix}3\\2\\6\end{pmatrix} = \begin{pmatrix}-3\\0\\2\end{pmatrix}, \qquad T\begin{pmatrix}1\\-1\\2\end{pmatrix} = \begin{pmatrix}2\\4\\-5\end{pmatrix}, \qquad T\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}2\\0\\5\end{pmatrix}?$$

WORKED SOLUTION

1. Test independence without determinants. If $au_1 + bu_2 + cu_3 = 0$, comparison of entries gives

$$a + 2b + c = 0, \qquad a - b = 0, \qquad a + 2b = 0.$$

The second equation gives $a = b$, and the third gives $3b = 0$. Thus $a = b = 0$, and then the first gives $c = 0$. Hence these vectors form a basis of $\mathbb{R}^3$. The theorem guarantees a unique linear map with the prescribed images.

To find it, first determine the action of T on the standard basis vectors e_1, e_2, e_3. Express

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EXERCISE 9.1 CONTINUED

each as a linear combination of the given vectors.

We have $e_1 = u_3$. For e_2, write $e_2 = au_1 + bu_2 + cu_3$. This gives a system for a, b, c.

Row reduction yields $a = \frac{2}{3}, b = -\frac{1}{3}, c = 0$.

Treating e_3 similarly gives

$$e_1 = u_3 \quad e_2 = \frac{2}{3}u_1 - \frac{1}{3}u_2, \quad e_3 = u_1 - e_1 - e_2$$

Apply linearity and the prescribed images:

$$T(e_1) = T(u_3) = e_1,$$

$$T(e_2) = \frac{2}{3}T(u_1) - \frac{1}{3}T(u_2) = \frac{2}{3}\begin{pmatrix}3\\0\\0\end{pmatrix} - \frac{1}{3}\begin{pmatrix}0\\0\\0\end{pmatrix} = \begin{pmatrix}2\\0\\0\end{pmatrix} = 2e_1,$$

$$T(e_3) = T(u_1) - T(e_1) - T(e_2) = 3e_1 - e_1 - 2e_1 = 0.$$

For every $v = (x, y, z)^T = xe_1 + ye_2 + ze_3$,

$$T(v) = xT(e_1) + yT(e_2) + zT(e_3) = xe_1 + 2ye_1 = \begin{pmatrix}x + 2y\\0\\0\end{pmatrix}.$$

Substituting the three original inputs verifies all required images.

2. Now set $u_1 = (3, 2, 6)^T$, $u_2 = (1, -1, 2)^T$, $u_3 = (0, 1, 0)^T$. These vectors are dependent, since

$$u_1 - 3u_2 - 5u_3 = (3 - 3,\ 2 + 3 - 5,\ 6 - 6)^T = 0.$$

Every linear map satisfies $T(0) = 0$. However, the prescribed images would give

$$T\begin{pmatrix}0\\0\\0\end{pmatrix} = T\left(\begin{pmatrix}3\\2\\6\end{pmatrix} - 3\begin{pmatrix}1\\-1\\2\end{pmatrix} - 5\begin{pmatrix}0\\1\\0\end{pmatrix}\right) =$$

$$T\begin{pmatrix}3\\2\\6\end{pmatrix} - 3T\begin{pmatrix}1\\-1\\2\end{pmatrix} - 5T\begin{pmatrix}0\\1\\0\end{pmatrix} =$$

$$\begin{pmatrix}-3\\0\\2\end{pmatrix} - 3\begin{pmatrix}2\\4\\-5\end{pmatrix} - 5\begin{pmatrix}2\\0\\5\end{pmatrix} = \begin{pmatrix}-19\\-12\\-8\end{pmatrix} \neq \begin{pmatrix}0\\0\\0\end{pmatrix}$$

so no such linear map exists.

A necessary distinction: One violated dependence relation suffices to disprove existence. To prove existence, all dependence relations among the inputs must be preserved by their prescribed images.

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MOTIVATION AND GUIDING IDEA

Existence criterion for prescribed images. The requirements $T(u_i) = w_i$ can be satisfied by a linear map if and only if, for every choice of coefficients a_i,

$$\sum_i a_i u_i = 0 \quad \Longrightarrow \quad \sum_i a_i w_i = 0.$$

Under this condition, define a map on $U = \operatorname{span}\{u_i\}$ by $T(\sum_i a_i u_i) = \sum_i a_i w_i$. The difference of two representations is a dependence relation, so both yield the same image. Linearity follows from the definition. If the domain is finite-dimensional, extend a basis of U to a basis of V, and freely prescribe images of the extra basis vectors. The map is unique if $U = V$. If $U \neq V$ and $W \neq \{0\}$, two different images may be chosen for an extra basis vector, so the extension is not unique. If the u_i already form a basis, there are no nontrivial dependence relations, and every choice of images defines a unique linear map.

EXERCISE 9.2

Let $V = \mathbb{R}^2$ and $W = \mathbb{R}_3[X]$. Find the unique linear map $T : V \to W$ satisfying

$$T((1,0)^T) = 2 + X^2, \qquad T((1,1)^T) = 1 + X + X^2.$$

Give an explicit formula for $T((x,y)^T)$.

WORKED SOLUTION

The vectors $(1,0)^T, (1,1)^T$ form a basis, since their column determinant is 1. Every vector has the unique expansion

$$(x,y)^T = (x-y)(1,0)^T + y(1,1)^T.$$

Linearity therefore requires

$$\begin{aligned} T((x,y)^T) &= (x-y)(2+X^2) + y(1+X+X^2) \\ &= 2x - 2y + (x-y)X^2 + y + yX + yX^2 \\ &= (2x-y) + yX + xX^2. \end{aligned}$$

The formula is linear because each polynomial coefficient is linear in x, y. The two given inputs produce $2 + X^2$ and $1 + X + X^2$, respectively. Existence and uniqueness follow from specifying the images of a basis.

9.2 Composition of maps

RECALL AND TOOLS

- Let U, V, W be vector spaces over a field $\mathbb{F}$.
- Let $T : U \to V$ and $S : V \to W$ be linear maps.
- The map $S \circ T : U \to W$, defined for $u \in U$ by $(S \circ T)(u) = S(T(u))$, is their **composition**.
- $S \circ T$ is linear over $\mathbb{F}$.

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- If both maps are surjective, their composition is surjective. If both are injective, their composition is injective. In particular, a composition of isomorphisms is an isomorphism.

EXERCISE 9.3

Compute $S \circ T$ for the following linear maps:
$T\colon \mathbb{R}^2 \longrightarrow \mathrm{M}_2(\mathbb{R})$ given by $T\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a & a+b \\ a-b & b \end{pmatrix}$
and $S : M_2(\mathbb{R}) \to \mathbb{R}_3[X]$, given by $S\begin{pmatrix} q & r \\ s & t \end{pmatrix} = tX^2 + (r+s)X + q$

WORKED SOLUTION

Compute $S \circ T : \mathbb{R}^2 \to \mathbb{R}_3[X]$. Let $\begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{R}^2$. Then

$$S \circ T \begin{pmatrix} a \\ b \end{pmatrix} = S(T\begin{pmatrix} a \\ b \end{pmatrix}) = S\begin{pmatrix} a & a+b \\ a-b & b \end{pmatrix} = bX^2 + 2aX + a$$

9.3 Representing a linear map by a matrix

DEFINITION 9.2

Let U be an n-dimensional vector space over $\mathbb{F}$ with ordered basis $B = (u_1, \ldots, u_n)$.
Let V be an m-dimensional vector space over the same field with ordered basis $C = (v_1, \ldots, v_m)$.
For a linear map $T : U \to V$, define the matrix

$$[T]_C^B = \begin{pmatrix} | & | & | & | \\ [T(u_1)]_C & [T(u_2)]_C & \cdots & [T(u_n)]_C \\ | & | & | & | \end{pmatrix} \in \mathbb{F}^{m \times n}$$

$[T]_C^B$ is the **matrix of T relative to the ordered bases B and C.**

Construction

First compute $T(u_i)$, the images of the basis vectors of the domain.

Then express each image in coordinates relative to C, and use these coordinate columns in the given order.

PROPOSITION 9.2

Let U, V be vector spaces over $\mathbb{F}$, with ordered bases B of U and C of V. For a linear map $T : U \to V$ and every $u \in U$,

$$[T(u)]_C = [T]_C^B \cdot [u]_B$$

Compute this matrix once for the chosen ordered bases. It then converts the coordinate vector of any new input into the coordinate vector of its image relative to C.

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EXERCISE 9.4

Consider the vector space

$$V = M_3(\mathbb{R})$$

of all real 3×3 matrices.
Define a linear map $T : V \to V$ by

$$T(A) = A - A^T.$$

Write

$$U = \{A \in V \mid A = A^T\}, \qquad W = \{A \in V \mid A = -A^T\}.$$

1. Prove that T is linear.
2. Find $\ker T$ and $\operatorname{Im} T$.
3. Show that $\ker T = U$ and $\operatorname{Im} T = W$.
4. Use rank-nullity to compute $\dim U$ and $\dim W$.
5. Find the matrix of T in an ordered basis obtained by concatenating a basis of U with a basis of W.

WORKED SOLUTION

1. For $A, B \in V$ and $\lambda \in \mathbb{R}$,

$$T(A+B) = (A+B) - (A+B)^T = (A - A^T) + (B - B^T) = T(A) + T(B),$$

and

$$T(\lambda A) = \lambda A - (\lambda A)^T = \lambda(A - A^T) = \lambda T(A),$$

so T is linear.

2. Begin with the kernel:

$$A \in \ker T \iff T(A) = 0 \iff A - A^T = 0 \iff A = A^T.$$

Therefore,

$$\ker T = U.$$

For the image, every $A \in V$ satisfies

$$T(A) = A - A^T,$$

and

$$(T(A))^T = (A - A^T)^T = A^T - A = -(A - A^T) = -T(A).$$

Thus $T(A) \in W$, so

$$\operatorname{Im} T \subseteq W.$$

Conversely, if $B \in W$, then $B = -B^T$. Set $A = \frac{1}{2}B$. Then

$$T(A) = \tfrac{1}{2}(B - B^T) = \tfrac{1}{2}(B - (-B)) = B,$$

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EXERCISE 9.4 CONTINUED

so $B \in \operatorname{Im} T$, proving

$$\operatorname{Im} T = W.$$

3. We have already shown that

$$\ker T = U, \qquad \operatorname{Im} T = W.$$

4. Apply rank-nullity. We know

$$\dim V = \dim M_3(\mathbb{R}) = 9.$$

and hence

$$\dim V = \dim \ker T + \dim \operatorname{Im} T = \dim U + \dim W.$$

A symmetric 3×3 matrix is determined by three diagonal entries and three entries above the diagonal. Thus

$$\dim U = 6.$$

Consequently,

$$\dim W = 9 - 6 = 3.$$

5. Choose the following basis of U:

$$E_{11},\ E_{22},\ E_{33},\ E_{12} + E_{21},\ E_{13} + E_{31},\ E_{23} + E_{32},$$

and the following basis of W:

$$E_{12} - E_{21},\ E_{13} - E_{31},\ E_{23} - E_{32}.$$

Form the ordered basis

$$B = (\text{basis of } U,\ \text{basis of } W).$$

Since $T(A) = 0$ for $A \in U$ and $T(B) = 2B$ for $B \in W$, the matrix of T in this ordered basis is

$$[T]_B = \begin{pmatrix} 0_{6\times 6} & 0 \\ 0 & 2I_3 \end{pmatrix}.$$

Thus T annihilates the U component and multiplies the W component by 2, in accordance with

$$V = U \oplus W.$$

EXERCISE 9.5

Let $V = \mathbb{R}^3$ with standard ordered basis $E = (e_1, e_2, e_3)$. Let $T : V \to V$ have the following matrix in E:

$$[T]_E = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix}.$$

1. Show that some nonzero $v \in V$ satisfies $T(v) = 0$.
2. Show that some nonzero $u \in V$ satisfies $T(u) = 3u$.

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EXERCISE 9.5 CONTINUED

3. Find $w \in V$ such that $T(w) = w + u$, using the u from part 2.

WORKED SOLUTION

Write $A = [T]_E$.

1. The first column of A is the coordinate vector of $T(e_1)$ relative to E. Here that column is zero, so
$$T(e_1) = 0.$$
Choose $v = e_1 \neq 0$; then $T(v) = 0$.

2. The third column of A is the coordinate vector of $T(e_3)$ relative to E. Here
$$T(e_3) = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} = e_1 - e_2 + 3e_3.$$
Seek $u = (x, y, z)$ with $T(u) = 3u$. Using A,
$$T(x, y, z) = A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} y + z \\ 2y - z \\ 3z \end{pmatrix}.$$
we require
$$\begin{pmatrix} y + z \\ 2y - z \\ 3z \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \\ 3z \end{pmatrix}.$$
The third equation, $3z = 3z$, is automatic. The second gives $2y - z = 3y$, or $y = -z$. The first gives $y + z = 3x$, so $x = 0$. Take $z = 1$, hence $y = -1$:
$$u = (0, -1, 1) \neq 0, \qquad T(u) = 3u.$$

3. From part 2, $u = (0, -1, 1)$. Seek $w = (x, y, z)$ satisfying $T(w) = w + u$. As above,
$$T(w) = \begin{pmatrix} y + z \\ 2y - z \\ 3z \end{pmatrix}, \qquad w + u = \begin{pmatrix} x \\ y - 1 \\ z + 1 \end{pmatrix}.$$
solve the system
$$\begin{cases} y + z = x, \\ 2y - z = y - 1, \\ 3z = z + 1. \end{cases}$$
The third equation gives $3z = z + 1$, hence $z = \frac{1}{2}$. The second gives $2y - z = y - 1$, hence $y = z - 1 = -\frac{1}{2}$. The first gives $x = y + z = 0$. Thus we may take
$$w = \left(0, -\tfrac{1}{2}, \tfrac{1}{2}\right), \qquad \text{and then} \qquad T(w) = w + u.$$

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EXERCISE 9.6

Let $V = \mathbb{C}[x]$, the vector space of all complex polynomials.
Define the linear operator $N : V \to V$ by

$$(Np)(x) := -(58 - x)\, p'(x).$$

For each $\theta \in \mathbb{R}$, define $U_\theta : V \to V$ by

$$U_\theta := \exp(-i\theta N^2) = \sum_{k=0}^{\infty} \frac{(-i\theta)^k}{k!} N^{2k}.$$

Use the following two ordered bases:

$$E = (1, x, x^2, x^3, \dots), \qquad B = (58,\ (58 - x),\ (58 - x)^2,\ (58 - x)^3, \dots).$$

1. Prove that N and U_θ are linear.
2. For $n \geq 1$, set $b_n(x) = (58 - x)^n$, and set $b_0(x) = 58$. Compute $N(b_n)$ and $N^2(b_n)$.
3. Compute $U_\theta(b_n)$ for all $n \geq 0$ and deduce the form of $[U_\theta]_B$.
4. Find the matrix of U_θ restricted to $P_3 = \text{span}\{1, x, x^2, x^3\}$ in the ordered basis $E_3 = (1, x, x^2, x^3)$.

WORKED SOLUTION

1. Differentiation is linear, so

$$N(\alpha p + \beta r) = -(58 - x)(\alpha p' + \beta r') = \alpha N(p) + \beta N(r).$$

To justify the series, restrict to P_d, the polynomials of degree at most d. This space is invariant under N, so on P_d the series is the exponential series of a finite matrix, converging for every θ. These definitions agree as d increases and therefore define a map on $\mathbb{C}[x]$. Linearity follows by taking limits of the partial sums in each coordinate.

2. For $n \geq 1$,

$$b_n'(x) = \frac{d}{dx}(58 - x)^n = n(58 - x)^{n-1} \cdot (-1) = -n(58 - x)^{n-1}.$$

so

$$N(b_n) = -(58 - x)\, b_n'(x) = -(58 - x)\big(-n(58 - x)^{n-1}\big) = n(58 - x)^n = nb_n.$$

and hence

$$N^2(b_n) = N(nb_n) = nN(b_n) = n \cdot nb_n = n^2 b_n.$$

Also $N(b_0) = N(58) = 0$, so $N^2(b_0) = 0$.

3. Since $N^{2k}(b_n) = n^{2k} b_n$ for $n \geq 1$, the series gives

$$U_\theta(b_n) = \sum_{k=0}^{\infty} \frac{(-i\theta)^k}{k!}\, n^{2k}\, b_n = \exp(-i\theta n^2)\, b_n.$$

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EXERCISE 9.6 CONTINUED

For $n = 0$, $U_\theta(b_0) = b_0$.

Thus the matrix in B is diagonal:

$$[U_\theta]_B = \operatorname{diag}\big(1,\ e^{-i\theta},\ e^{-i4\theta},\ e^{-i9\theta}, \dots\big).$$

4. Write $q_1 = e^{-i\theta}$, $q_4 = e^{-i4\theta}$ and $q_9 = e^{-i9\theta}$. On P_3, use b_0, b_1, b_2, b_3. Their images are $b_0, q_1 b_1, q_4 b_2, q_9 b_3$, respectively.

   Expand these four polynomials in E_3:

$$b_0 = 58, \qquad b_1 = 58 - x, \qquad b_2 = (58 - x)^2 = x^2 - 116x + 3364,$$

$$b_3 = (58 - x)^3 = -x^3 + 174x^2 - 10092x + 195112.$$

   Let C be the matrix with coordinate columns $[b_0]_{E_3}, [b_1]_{E_3}, [b_2]_{E_3}, [b_3]_{E_3}$:

$$C = \begin{pmatrix} 58 & 58 & 3364 & 195112 \\ 0 & -1 & -116 & -10092 \\ 0 & 0 & 1 & 174 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

   This matrix is invertible. The formula $[U_\theta]_{E_3} = C\,\operatorname{diag}(1, q_1, q_4, q_9)C^{-1}$ gives the following matrix:

$$[U_\theta]_{E_3} = \begin{pmatrix} 1 & 58(1-q_1) & 3364(1-2q_1+q_4) & 195112(1-3q_1+3q_4-q_9) \\ 0 & q_1 & 116(q_1-q_4) & 10092(q_1-2q_4+q_9) \\ 0 & 0 & q_4 & 174(q_4-q_9) \\ 0 & 0 & 0 & q_9 \end{pmatrix}.$$

EXERCISE 9.7

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be given by

$$T((x, y, z)^T) = (2x - y + z,\ x + 3z)^T.$$

Find its matrix in the standard ordered bases by computing the images of the three basis vectors.

WORKED SOLUTION

Substitute each basis vector:

$$T(e_1) = (2 \cdot 1 - 0 + 0,\ 1 + 3 \cdot 0)^T = (2, 1)^T,$$

$$T(e_2) = (2 \cdot 0 - 1 + 0,\ 0 + 3 \cdot 0)^T = (-1, 0)^T,$$

$$T(e_3) = (2 \cdot 0 - 0 + 1,\ 0 + 3 \cdot 1)^T = (1, 3)^T.$$

These images, in this order, are the columns of the matrix:

$$[T]_{E_2}^{E_3} = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 0 & 3 \end{pmatrix}.$$

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EXERCISE 9.7 CONTINUED

As a check, multiplying by a general vector yields $(2x-y+z, x+3z)^T$, the stated formula.

EXERCISE 9.8

Let $T:\mathbb{R}_2[X]\to\mathbb{R}^2$ be the linear map given by $T(P(X))=\begin{pmatrix}P(1)\\P(2)\end{pmatrix}$ for every $P(X)\in\mathbb{R}_2[X]$.

Let $B=\left(1,1+X,1+X+X^2\right)$ and $C=\left(\begin{pmatrix}1\\1\end{pmatrix},\begin{pmatrix}0\\1\end{pmatrix}\right)$ be ordered bases of $\mathbb{R}_2[X]$ and $\mathbb{R}^2$, respectively.

(a) Find $[T]_C^B$. (b) **(Independent practice)** Verify the identity from the preceding theorem, namely $[T(u)]_C=[T]_C^B[u]_B$

WORKED SOLUTION

By definition, compute $[T(1)]_C, [T(1+X)]_C, [T(1+X+X^2)]_C$:

$$[T(1)]_C=[\begin{pmatrix}P(1)=1\\P(2)=1\end{pmatrix}]_C=[\begin{pmatrix}1\\1\end{pmatrix}]_C=[1\cdot\begin{pmatrix}1\\1\end{pmatrix}+0\cdot\begin{pmatrix}0\\1\end{pmatrix}]_C=\begin{pmatrix}1\\0\end{pmatrix}$$

$$[T(1+X)]_C=[\begin{pmatrix}P(1)=2\\P(2)=3\end{pmatrix}]_C=[\begin{pmatrix}2\\3\end{pmatrix}]_C=[2\cdot\begin{pmatrix}1\\1\end{pmatrix}+1\cdot\begin{pmatrix}0\\1\end{pmatrix}]_C=\begin{pmatrix}2\\1\end{pmatrix}$$

$$[T(1+X+X^2)]_C=[\begin{pmatrix}P(1)=3\\P(2)=7\end{pmatrix}]_C=[\begin{pmatrix}3\\7\end{pmatrix}]_C=[3\cdot\begin{pmatrix}1\\1\end{pmatrix}+4\cdot\begin{pmatrix}0\\1\end{pmatrix}]_C=\begin{pmatrix}3\\4\end{pmatrix}$$

We obtain $[T]_C^B=\begin{pmatrix}1&2&3\\0&1&4\end{pmatrix}$.

(b) To verify $[T(u)]_C=[T]_C^B[u]_B$, take a general polynomial $P(X)=aX^2+bX+c\in\mathbb{R}_2[X]$. We check the equality for this polynomial.

First compute the left-hand side:

$$[T(P(X))]_C=[\begin{pmatrix}P(1)=a+b+c\\P(2)=4a+2b+c\end{pmatrix}]_C=[(a+b+c)\begin{pmatrix}1\\1\end{pmatrix}+(3a+b)\begin{pmatrix}0\\1\end{pmatrix}]_C=\begin{pmatrix}a+b+c\\3a+b\end{pmatrix}$$

For the right-hand side, find the coordinates of $P(X)$ relative to B: solve for $\alpha_1,\alpha_2,\alpha_3$ such that

$$aX^2+bX+c=\alpha_1\cdot 1+\alpha_2\cdot(1+X)+\alpha_3\cdot(1+X+X^2)$$

Comparison of coefficients, or row reduction, gives $\alpha_1=c-b, \alpha_2=b-a, \alpha_3=a$.

Thus $[P(X)]_B=\begin{pmatrix}c-b\\b-a\\a\end{pmatrix}$.

Using the result of part (a), the product is equal to the previously computed left-hand side:

$$[T]_C^B[P(X)]_B=\begin{pmatrix}1&2&3\\0&1&4\end{pmatrix}\begin{pmatrix}c-b\\b-a\\a\end{pmatrix}=\begin{pmatrix}c-b+2(b-a)+3a\\b-a+4a\end{pmatrix}=\begin{pmatrix}a+b+c\\3a+b\end{pmatrix}$$

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9.3.1 Change-of-coordinate matrices

RECALL AND TOOLS

Definition: Let V be a finite-dimensional vector space over $\mathbb{F}$, with ordered bases B, B'. The **change-of-coordinate matrix** $[I_V]_B^{B'}$ from B' to B is the representing matrix $[T]_B^{B'}$ for the **identity map** $T : V \to V$, $T(v) = v$ for every $v \in V$.

- The fundamental identity is $[v]_B = [I_V]_B^{B'}[v]_{B'}$.
- The identity map is denoted I_V, with $I_V(v) = v$.
- Its columns are the coordinate vectors of the vectors of B' relative to the ordered basis B.

EXERCISE 9.9

In $V = \mathbb{R}_1[X]$, consider the ordered bases

$$B = (1 + X,\ 1 + 2X), \qquad B' = (2 + 3X,\ 3 + 5X).$$

Compute $[I_V]_B^{B'}$, converting coordinates from B' to B. For $p(X) = a + bX$, explicitly verify $[p]_B = [I_V]_B^{B'}[p]_{B'}$.

WORKED SOLUTION

The basis-column matrices in the standard ordered basis $(1, X)$ are

$$M_B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad M_{B'} = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}.$$

Both have determinant 1. For the first change-of-coordinate column, use $2 + 3X = (1 + X) + (1 + 2X)$; for the second, use $3 + 5X = (1 + X) + 2(1 + 2X)$. Thus

$$P = [I_V]_B^{B'} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

For a general polynomial, solve for its coordinates in each basis:

$$\alpha + \beta = a,\ \alpha + 2\beta = b \quad \Longrightarrow \quad [p]_B = (2a - b, b - a)^T,$$

$$2\gamma + 3\delta = a,\ 3\gamma + 5\delta = b \quad \Longrightarrow \quad [p]_{B'} = (5a - 3b, -3a + 2b)^T.$$

Now compute both entries of the product:

$$P[p]_{B'} = \begin{pmatrix} (5a - 3b) + (-3a + 2b) \\ (5a - 3b) + 2(-3a + 2b) \end{pmatrix} = \begin{pmatrix} 2a - b \\ b - a \end{pmatrix} = [p]_B.$$

This also verifies the direction: B'-coordinates are the input, and B-coordinates are the output.

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9.3.2 Matrices of compositions and inverse maps

PROPOSITION 9.3

Let U, V, W be finite-dimensional vector spaces over $\mathbb{F}$, with ordered bases A, B, C, respectively.
Let $S : U \to V$ and $T : V \to W$ be linear. Then

$$[T \circ S]_C^A = [T]_C^B [S]_B^A$$

Corollary

If $T : V \to W$ is an **isomorphism** and B, C are ordered bases of V, W, respectively, then

$$[T^{-1}]_B^C = \left([T]_C^B\right)^{-1}$$

9.4 Similar matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are similar if there is an invertible $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

1. $$\det(A) = \det(B)$$
2. $$\operatorname{trace}(A) = \operatorname{trace}(B)$$
3. $$\operatorname{rank}(A) = \operatorname{rank}(B)$$

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EXERCISE 9.10

Let $A, B \in \mathbb{F}^{n\times n}$. Prove or disprove:

1. If A, B are similar, then A^k, B^k are similar for every integer $k \geq 1$.
2. If A^k, B^k are similar for some integer $k \geq 1$, then A, B are similar.

WORKED SOLUTION

1. If $B = P^{-1}AP$ with P invertible, cancellation of $PP^{-1} = I$ gives

$$B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

Thus the powers are similar via the same P.

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EXERCISE 9.10 CONTINUED

2. The claim is false. Over any field, take

$$A = 0_{2\times 2}, \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $A^2 = B^2 = 0$, but A, B are not similar: $P^{-1}AP = 0$ for every invertible P, whereas $B \neq 0$. Over $\mathbb{R}$ or $\mathbb{C}$, one may also take $A = -I_2, B = I_2$: their squares agree, and a scalar matrix is similar only to itself. This example does not disprove the statement in characteristic 2, where $-I_2 = I_2$.

Common error: The identity $A^k = (Q^{-1}BQ)^k$ does not imply $A = Q^{-1}BQ$. Taking matrix powers is not an injective operation, as these examples show.

EXERCISE 9.11

A planar system represents a force vector $v = (x, y)^T$ by two sensor readings:

$$T(v) = \begin{pmatrix} x + y \\ x - y \end{pmatrix}.$$

Use $B = ((1, 1)^T, (1, -1)^T)$ for input coordinates and the standard basis E for output coordinates. Find $[T]_E^B$, recover v from $T(v) = (6, 2)^T$, and check the answer in both coordinate systems.

WORKED SOLUTION

The columns are the images of the input basis vectors:

$$T(1, 1)^T = (2, 0)^T, \qquad T(1, -1)^T = (0, 2)^T, \qquad [T]_E^B = 2I_2.$$

Thus if $[v]_B = (a, b)^T$, then $(6, 2)^T = 2(a, b)^T$, giving $a = 3, b = 1$. Hence

$$v = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}.$$

In standard coordinates, $T((4, 2)^T) = (4 + 2, 4 - 2)^T = (6, 2)^T$. Using B for the input and E for the output gives $[T]_E^B = 2I_2$, so the two coordinate channels are uncoupled. This uses different bases for the domain and codomain; it does not assert that $[T]_B^B = 2I_2$.

EXERCISE 9.12 EXAM QUESTION

Prove or disprove the following statements.

1. There do not exist fields $\mathbb{F}_1 \subseteq \mathbb{F}_2$ and a vector space V over $\mathbb{F}_2$, also regarded as an $\mathbb{F}_1$-vector space by restricting scalars, such that

$$\dim_{\mathbb{F}_1} V = 2 \dim_{\mathbb{F}_2} V.$$

2. If $A \in M_n(\mathbb{F})$ is invertible, then every $B \in M_{n\times m}(\mathbb{F})$ satisfies

$$N(AB) = N(B).$$

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EXERCISE 9.12 EXAM QUESTION CONTINUED

3. Let $n \geq 1$. For every integer k with $0 \leq k \leq n^2$, there exists $A \in M_n(\mathbb{R})$ with $\operatorname{tr}(A) = k$ that is similar only to itself.

WORKED SOLUTION

1. False. Take $\mathbb{F}_1 = \mathbb{R}$, $\mathbb{F}_2 = \mathbb{C}$ and $V = \mathbb{C}$. Over $\mathbb{C}$, the basis (1) gives

$$\dim_{\mathbb{C}} \mathbb{C} = 1.$$

Over $\mathbb{R}$, the basis $(1, i)$ gives

$$\dim_{\mathbb{R}} \mathbb{C} = 2 = 2 \dim_{\mathbb{C}} \mathbb{C}.$$

The underlying vectors are the same, but the dimension changes because the field of admissible coefficients changes.

2. True. Prove both inclusions. If $x \in N(B)$, then $Bx = 0$, and

$$ABx = A(Bx) = A0 = 0.$$

so $x \in N(AB)$. Hence $N(B) \subseteq N(AB)$. Conversely, if $x \in N(AB)$, then $ABx = 0$. Multiplication on the left by A^{-1} gives

$$Bx = (A^{-1}A)Bx = A^{-1}0 = 0.$$

so $x \in N(B)$, proving $N(AB) \subseteq N(B)$. Together the inclusions give the required equality.

3. True. Define

$$A = \frac{k}{n} I_n.$$

Since $n \geq 1$ and the field is $\mathbb{R}$, the quotient is defined, and

$$\operatorname{tr}(A) = \sum_{j=1}^{n} \frac{k}{n} = n\frac{k}{n} = k.$$

For every invertible $P \in M_n(\mathbb{R})$,

$$P^{-1}AP = P^{-1}\left(\frac{k}{n} I_n\right) P = \frac{k}{n} P^{-1}P = \frac{k}{n} I_n = A.$$

Thus every matrix similar to A equals A. In fact this construction works for every real k, including all the values in the statement.

Additional practice

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EXERCISE 9.13 FOUNDATION

Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(x, y) = (2x + y, x + y)$, and choose the ordered bases

$$B = ((1, 0)^T, (1, 1)^T), \qquad C = ((1, 1)^T, (0, 1)^T).$$

Compute $[T]_C^B$ and $[T^{-1}]_B^C$. For $v = (3, 1)^T$, verify the coordinate identity and recover v using the inverse.

WORKED SOLUTION

For every $(r, s)^T$, its C-coordinates are $(r, s - r)^T$. Since $T((1, 0)^T) = (2, 1)^T$ and $T((1, 1)^T) = (3, 2)^T$,

$$M = [T]_C^B = \begin{pmatrix} 2 & 3 \\ -1 & -1 \end{pmatrix}, \qquad \det M = 1.$$

Thus T is an isomorphism. Reversing the roles of the domain and codomain bases gives

$$[T^{-1}]_B^C = M^{-1} = \begin{pmatrix} -1 & -3 \\ 1 & 2 \end{pmatrix}.$$

Now $v = 2(1, 0)^T + (1, 1)^T$, so $[v]_B = (2, 1)^T$. Direct computation gives $T(v) = (7, 4)^T$, and hence

$$[T(v)]_C = \begin{pmatrix} 7 \\ -3 \end{pmatrix} = M \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Finally, $M^{-1}(7, -3)^T = (2, 1)^T$, and reconstruction in B gives $(3, 1)^T$. The inverse matrix takes C-coordinates as input and returns B-coordinates.

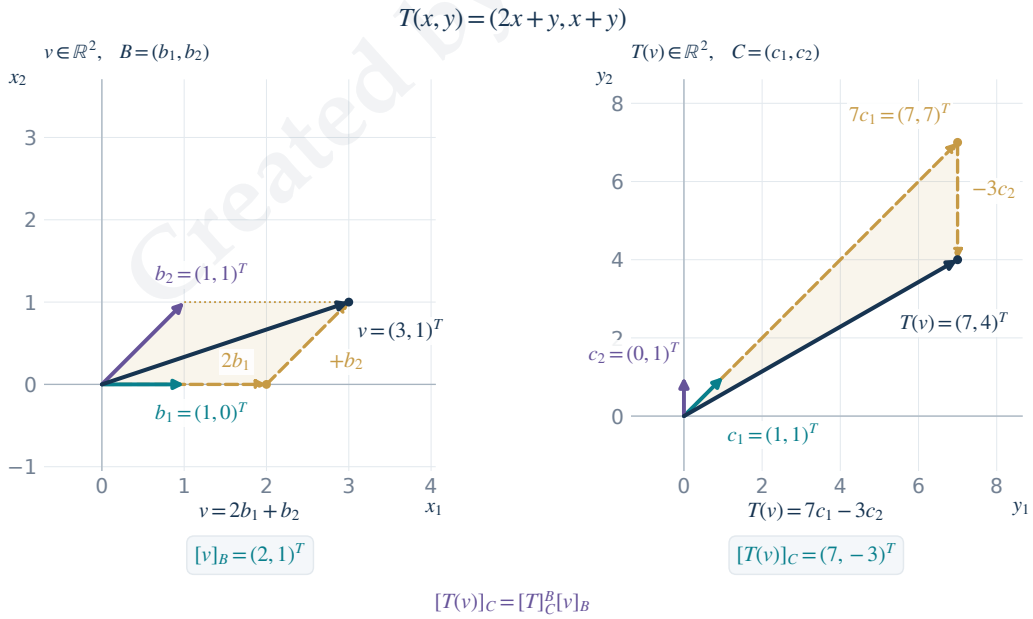


Figure 9.1. The arrows show vectors in standard coordinates; the gold paths show the decompositions $v = 2b_1 + b_2$ and $T(v) = 7c_1 - 3c_2$. Coordinate columns appear separately: $[v]_B = (2, 1)^T$ and $[T(v)]_C = (7, -3)^T$. The vector $T(v)$ itself is $(7, 4)^T$.

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EXERCISE 9.14 CHALLENGE

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ have standard matrix

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

and let $B = ((1,1)^T, (1,-1)^T)$. Compute $D = [T]_B$ and $P = [I]_E^B$. Prove $A^k = PD^kP^{-1}$ for every integer $k \geq 0$, and use it to solve $x_{k+1} = Ax_k$ with $x_0 = (1,0)^T$.

WORKED SOLUTION

The basis images are $T((1,1)^T) = 4(1,1)^T$ and $T((1,-1)^T) = 2(1,-1)^T$. Therefore,

$$D = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The columns of P reconstruct standard coordinates from B-coefficients. Hence $D = P^{-1}AP$ and $A = PDP^{-1}$. For $k = 0$, the identity is $I = PIP^{-1}$. For $k \geq 1$, cancelling adjacent factors $P^{-1}P$ gives $A^k = PD^kP^{-1}$. Explicitly,

$$A^k = \frac{1}{2}\begin{pmatrix} 4^k + 2^k & 4^k - 2^k \\ 4^k - 2^k & 4^k + 2^k \end{pmatrix}.$$

Induction on the recurrence gives $x_k = A^k x_0$, so

$$x_k = \frac{1}{2}\begin{pmatrix} 4^k + 2^k \\ 4^k - 2^k \end{pmatrix} \qquad (k \geq 0).$$

At $k = 0$ this is $(1,0)^T$; for $k \geq 1$ both entries are integers. In B-coordinates the components evolve separately by multiplication by 4 and 2, which explains the simpler calculation.

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